CAMBRIDGE
International Education

Cambridge IGCSE™

PHYSICS **0625/52**

Paper 5 Practical **February/March 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1: Marks must be awarded in line with: • the specific content of the mark scheme or the generic level descriptors for the question • the specific skills defined in the mark scheme or in the generic level descriptors for the question • the standard of response required by a candidate as exemplified by the standardisation scripts.
GENERIC MARKING PRINCIPLE 2: Marks awarded are always **whole marks** (not half marks, or other fractions).
GENERIC MARKING PRINCIPLE 3: Marks must be awarded **positively**: • marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate • marks are awarded when candidates clearly demonstrate what they know and can do • marks are not deducted for errors • marks are not deducted for omissions • answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.
GENERIC MARKING PRINCIPLE 4: Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>‘List rule’ guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
✓	correct point or mark awarded
✗	incorrect point or mark not awarded
^	information missing or insufficient for credit
A	allow or accept
AE	arithmetic error
I	incorrect or insufficient point ignored while marking the rest of the response
CON	contradiction in response, mark not awarded
BOD	benefit of the doubt given
ECF	error carried forward applied
NAQ	response has not answered question

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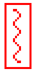
Annotation	Meaning
RE	rounding error
SEEN or 	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
TE	transcription error
TV	response is too vague or there is insufficient detail in response
T	answer outside the tolerance of the mark scheme
	used to highlight parts of an extended response
	used to highlight parts of an extended response
MO	mandatory mark not awarded
SC	special case
	unclear response
PD	poor diagram
POT	power of ten error
XP	incorrect physics
U	incorrect unit

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Question	Answer	Marks
1(a)(i)	value in range 50 to 500, to nearest g ;	**1**
1(a)(ii)	any **one** from: zero / tare balance before measurement ; subtract mass reading from before cylinder placed on balance ; place cylinder on centre of balance ;	**1**
1(a)(iii)	value in range 25 to 75, to nearest ml ;	**1**
1(a)(iv)	any **one** from: measure from bottom of meniscus ; measure with eye level with water surface / meniscus ; perpendicular viewing / at a right angle ;	**1**
1(b)(i)	$m_2 > m_1$;	**1**
1(b)(ii)	$(m_2 – m_1) / 5$	**1**
1(b)(iii)	$V_2 > V_1$, to nearest ml ;	**1**
1(b)(iv)	$(V_2 – V_1) / 5$	**1**
1(b)(v)	correct calculation of ρ using their **1(b)(ii)** and **1(b)(iv)**;	**1**
1(c)	any **two** from: measuring cylinder with smaller scale divisions ; measure diameter / radius of ball and calculate volume ; use more glass balls (so there is a greater change in volume) ; repeat and calculate mean ;	**2**

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Question	Answer	Marks
2(a)(i)	value in range 15–25 mm, in mm to the nearest mm ;	**1**
2(a)(ii)	diagram of spring, clearly showing l_0 spanning coiled section of spring and not including loops at end of spring	**1**
2(a)(iii)	any **two** from: ruler close to spring ruler parallel to spring / ruler vertical eye level with reading / read perpendicular use marker / pointer / set square / horizontal rulers to align spring with ruler	**2**
2(b)	increasing values of l ;	**1**
	correct calculation of *x* values ;	**1**
2(c)(i)	correct labels and units: *L* / N (y-axis) against *x* / mm (x-axis) ;	**1**
	suitable scale filling more than half the graph grid in both directions, beginning from origin ;	**1**
	five plots located within half a small square ;	**1**
	line of best fit with balance of points, and single, thin continuous line ;	**1**
2(c)(ii)	if candidate line within 1 small square of origin: yes, (because line is) <u>straight</u> line through origin ; otherwise: no, (because line is) not a <u>straight</u> line / not through origin ;	**1**

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Question	Answer	Marks
3(a)(i)	All three V less than 4.0 V and all three I less than 1.00 A ;	1
3(a)(ii)	units V / V and I / A ;	1
3(a)(iii)	correct calculation of R for both circuits and consistent 2 or consistent 3 sf ;	1
	both R within 20% of each other ;	1
3(a)(iv)	statement matching results;	1
	if yes: within limits of experimental accuracy / within 10% / the values are the same / the value are close ; if no: not within limits of experimental accuracy / greater than 10% difference / / the value far apart ;	1
3(a)(v)	value equal to either of their *R* values or between their 2 *R* values or X ;	1
3(b)(i)	all added components in series ;	1
	three resistors with correct circuit symbols ;	1
3(b)(ii)	All given V recorded to at least 1dp and all given I recoded to at least 2dp ;	1
3(c)	any **one** from: **to prevent**: circuit becoming warm / overheating / heating of wires / heating of resistors / battery discharging ;	1

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td rowspan="6">4</td><td>MP1 apparatus:
thermometer <u>and</u> stop-watch</td><td>1</td></tr>
<tr><td>MP2 method:
measure starting temperature and temperature after given time
OR
measure starting temperature and time to drop to given temperature / room temp</td><td>1</td></tr>
<tr><td>MP3 repeat for different starting temperatures</td><td>1</td></tr>
<tr><td>MP4 control variable:
constant volume of water or control variable is volume of water</td><td>1</td></tr>
<tr><td>MP5 table:
columns, with units, for starting temperature and dependent variable
<table>
<tr><td>starting temperature / °C</td><td>final temperature / °C</td></tr>
<tr><td></td><td></td></tr>
</table>
OR
<table>
<tr><td>starting temperature / °C</td><td>time (to drop to given temperature) / s</td></tr>
<tr><td></td><td></td></tr>
</table>
</td><td>1</td></tr>
<tr><td>MP6 analysis:
any one from:
compare readings to see if change in initial temperature produces change in dependent variable
compare readings to see if change in initial temperature produces change in rate of cooling
plot (line) graph (with axes specified)</td><td>1</td></tr>
</table>

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Question	Answer	Marks
4	MP7 **additional point:** any **one** from: • at least 5 starting temperatures used • repeat for each value of independent variable and take average • room temperature constant • description of calculating rate of cooling from results • use lid every time	**1**

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